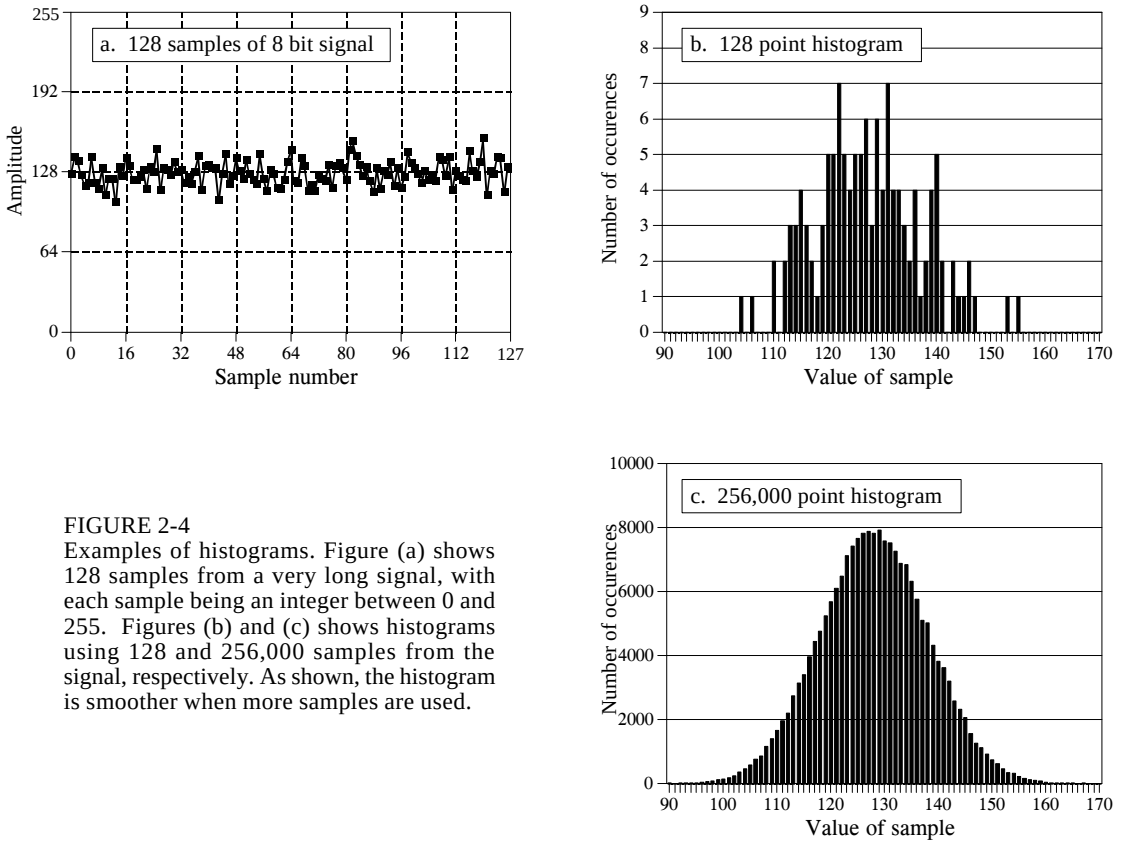

FIGURE 2-4
Examples of histograms. Figure (a) shows 128 samples from a very long signal, with each sample being an integer between 0 and 255. Figures (b) and (c) shows histograms using 128 and 256,000 samples from the signal, respectively. As shown, the histogram is smoother when more samples are used.

example, there are 2 samples that have a value of 110, 8 samples that have a value of 131, 0 samples that have a value of 170, etc. We will represent the histogram by H_i, where i is an index that runs from 0 to M-1, and M is the number of possible values that each sample can take on. For instance, H_{50} is the number of samples that have a value of 50. Figure (c) shows the histogram of the signal using the full data set, all 256k points. As can be seen, the larger number of samples results in a much smoother appearance. Just as with the mean, the statistical noise (roughness) of the histogram is inversely proportional to the square root of the number of samples used.

From the way it is defined, the sum of all of the values in the histogram must be equal to the number of points in the signal:

EQUATION 2-5
The sum of all of the values in the histogram is equal to the number of points in the signal. In this equation, H_i is the histogram, N is the number of points in the signal, and M is the number of points in the histogram.

$$N = \sum_{i=0}^{M-1} H_i$$

The histogram can be used to efficiently calculate the mean and standard deviation of very large data sets. This is especially important for *images*, which can contain millions of samples. The histogram groups samples